

Phase 6: Windows Client Machine

Deploying a Windows client workstation and joining it to the Active Directory domain.

Create CLIENT1

Using VirtualBox, create a new virtual machine named:

CLIENT1

Recommended specifications:

- Operating System: Windows 11 Pro
- Memory: 4096 MB
- Processors: 2
- Storage: 60 GB dynamically allocated
- Network Adapter: Bridged Adapter

Install Windows 11 Pro

After downloading the Windows Installation Media Tool from <https://www.microsoft.com/en-us/software-download/windows11>, Install Windows 11 Pro normally.

Create a local administrator account during setup. This account can be used later for troubleshooting and local administration.

Verify Network Connectivity

Open Command Prompt and run:

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ipconfig
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Verify that CLIENT1 has received an IP address on the same subnet as DC1.

Configure DNS

Open:

Settings > Network & Internet > Ethernet > Edit DNS Server Assignment

Configure the Preferred DNS Server as:

192.168.1.10

Leave the Alternate DNS Server blank.

Active Directory relies heavily on DNS. Proper DNS configuration is essential for successful domain communication.

Verify Name Resolution

Open Command Prompt and run:

nslookup

Verify that the default server points to:

DC1.crittenden.lab

Address:

192.168.1.10

Run the following commands to verify connectivity:

- nslookup dc1
- nslookup crittenden.lab
- ping dc1
- ping crittenden.lab

Successful responses confirm that DNS and network communication are functioning correctly.

Join CLIENT1 to the Domain

Press:

Win + R

Type:

sysdm.cpl

Select the Computer Name tab and click Change.

Under Member Of, select:

Domain

Enter:

crittenden.lab

When prompted, enter domain administrator credentials.

Upon successful authentication, a message welcoming CLIENT1 to the domain should appear.

Restart the computer.

Verify the Computer Account

On DC1, open:

Tools > Active Directory Users and Computers

Locate:

CLIENT1

inside the Computers container.

Move CLIENT1 into the Computers Organizational Unit to maintain proper organization and prepare for future Group Policy assignments.

Log In Using a Domain Account

At the Windows sign-in screen, select:

Other User

Sign in using a domain account.

Examples:

CRITTENDEN\jsmith

or

jsmith@crittenden.lab

Verify Domain Authentication

Open Command Prompt and run:

whoami

Successful authentication should return:

crittenden\username

This confirms that the workstation is authenticating against Active Directory rather than using a local account.

Troubleshooting Notes

During the initial deployment of CLIENT1, Windows 11 Home was installed. It was discovered that Windows 11 Home does not support Active Directory domain membership.

After identifying the issue, CLIENT1 was rebuilt using Windows 11 Pro. Once the correct operating system edition was installed, domain connectivity and domain join operations completed successfully.

This experience reinforced the importance of verifying operating system requirements and demonstrated the value of troubleshooting during system administration tasks.